AJ-ORBIT CIRCUIT INTERCONNECT

Full Hardware Wiring, Pin Mapping & Power Architecture

SCHEMATIC REF: V1.0

1. L298N MOTOR DRIVER ↔ ARDUINO UNO LOGIC INTERCONNECT

DRIVER TERMINAL	ARDUINO UNO PIN	SIGNAL MODE	FUNCTIONAL DESCRIPTION	WIRE CODE
ENA (Enable A)	Digital Pin 11	PWM	Controls Left Motor speed (BASE_SPEED / TURN_SPEED)	Yellow
IN1	Digital Pin 10	DIGITAL	Left Motor Direction Control (Phase 1)	Blue
IN2	Digital Pin 9	DIGITAL	Left Motor Direction Control (Phase 2)	Green
ENB (Enable B)	Digital Pin 6	PWM	Controls Right Motor speed (BASE_SPEED / TURN_SPEED)	Yellow
IN3	Digital Pin 8	DIGITAL	Right Motor Direction Control (Phase 1)	Blue
IN4	Digital Pin 7	DIGITAL	Right Motor Direction Control (Phase 2)	Green

2. DUAL IR LINE TRACKING SENSOR MODULES ↔ ARDUINO UNO

SENSOR MODULE PIN	ARDUINO UNO PIN	SIGNAL MODE	FUNCTIONAL DESCRIPTION	WIRE CODE
Left Sensor OUT	Digital Pin 2	INPUT	Left reflection feedback (LOW = White, HIGH = Black)	White
Left Sensor VCC	5V Power Rail	POWER	Regulated 5V sensor supply from Arduino header	Red
Left Sensor GND	GND Pin	GROUND	Common digital logic ground reference	Black
Right Sensor OUT	Digital Pin 3	INPUT	Right reflection feedback (LOW = White, HIGH = Black)	White
Right Sensor VCC	5V Power Rail	POWER	Regulated 5V sensor supply from Arduino header	Red
Right Sensor GND	GND Pin	GROUND	Common digital logic ground reference	Black

3. POWER SUPPLY, MOTORS & COMMON GROUND ARCHITECTURE

SOURCE / COMPONENT	TARGET TERMINAL	VOLTAGE / NET	SAFETY & WIRING NOTES
2×18650 Pack (+)	L298N 12V Screw Port	7.4V BATT	Pass through SPST toggle switch on wood chassis for main power cut.
2×18650 Pack (−)	L298N GND Screw Port	SYS GND	Tie battery negative directly to driver power ground terminal.
L298N GND Port	Arduino GND Pin	COMMON	**MANDATORY:** Common ground link prevents erratic floating PWM signals.
Motor A (Left)	OUT1 & OUT2	H-BRIDGE	Reversible polarity DC output; swap leads if wheel turns backwards.
Motor B (Right)	OUT3 & OUT4	H-BRIDGE	Reversible polarity DC output; swap leads if wheel turns backwards.

JUMPER CAP NOTICE

Keep the **5V Enable Jumper** plugged on the L298N board when powering via 7.4V to use its onboard 5V regulator for logic. Remove it if supplying >12V.

ZERO-FLOAT RULE

Never leave the Arduino GND disconnected from the L298N GND. A missing common ground reference will cause motor jitter and false IR readings.